

27 Supersymmetric Gauge Theories

27.2 Gauge-Invariant Action for Abelian Gauge Superfields

We must now consider how to construct a gauge-invariant supersymmetric action for the gauge superfields $V^A(x, \theta)$ that contain the gauge fields $V_\mu^A(x)$. In order to motivate this construction we will first consider the case of a single Abelian gauge field (dropping the superscript A), and then return to the general case in the next section.

In an Abelian gauge theory like quantum electrodynamics, the gauge-invariant field constructed from $V_\mu(x)$ is the familiar field-strength tensor

$$f_{\mu\nu}(x) = \partial_\mu V_\nu(x) - \partial_\nu V_\mu(x). \quad (27.2.1)$$

The supersymmetry transformation rule for $f_{\mu\nu}(x)$ is then given by the transformation rule (26.2.15) for $V_\mu(x)$ as

$$\delta f_{\mu\nu} = (\bar{\alpha}(\partial_\mu \gamma_\nu - \partial_\nu \gamma_\mu)\lambda). \quad (27.2.2)$$

Eq. (26.2.16) gives the transformation rule for $\lambda(x)$ as

$$\delta\lambda = \left(-\frac{1}{4}f_{\mu\nu}[\gamma^\mu, \gamma^\nu] + i\gamma_5 D \right) \alpha, \quad (27.2.3)$$

while Eq. (26.2.17) gives the transformation rule for $D(x)$:

$$\delta D = i(\bar{\alpha}\gamma_5 \not{\partial}\lambda). \quad (27.2.4)$$

None of this depends on whether or not the superfield $V^A(x, \theta)$ is taken to be in Wess–Zumino gauge. We see that the fields $f_{\mu\nu}(x)$, $\lambda(x)$, and $D(x)$ form a complete supersymmetry multiplet.

It is not hard to construct a suitable kinematic Lagrangian density for the fields of this supermultiplet. The only Lorentz-invariant, parity conserving, and gauge-invariant functions of these fields with dimensionality four are $f_{\mu\nu}f^{\mu\nu}$, $\bar{\lambda}\not{\partial}\lambda$, and D^2 . We can make V^μ a conventionally normalized vector field by taking the coefficient of $f_{\mu\nu}f^{\mu\nu}$ to be $-1/4$, so we may tentatively take the kinematic Lagrangian density as

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} - c_\lambda(\bar{\lambda}\not{\partial}\lambda) - c_D D^2,$$

with coefficients c_λ and c_D to be determined from the condition that $\int \mathcal{L}_{\text{gauge}} d^4x$ is supersymmetric. Using Eqs. (27.2.2)–(27.2.4), an infinitesimal supersymmetry transformation changes the operators in the Lagrangian by

$$\begin{aligned} \delta(f_{\mu\nu}f^{\mu\nu}) &= 2f^{\mu\nu}(\bar{\alpha}(\gamma_\nu\partial_\mu - \gamma_\mu\partial_\nu)\lambda), \\ \delta(\bar{\lambda}\not{\partial}\lambda) &= 2\left(\bar{\alpha}\left[-\frac{1}{4}f_{\mu\nu}[\gamma^\mu, \gamma^\nu] + i\gamma_5 D\right]\not{\partial}\lambda\right), \\ \delta D^2 &= 2iD(\bar{\alpha}\gamma_5 \not{\partial}\lambda), \end{aligned}$$

in which we drop derivative terms that do not contribute to the variation of the action. In order to see how these terms cancel, it is necessary to use an identity for gamma matrices¹

$$[\gamma^\mu, \gamma^\nu]\gamma^\rho = -2\eta^{\mu\rho}\gamma^\nu + 2\eta^{\nu\rho}\gamma^\mu - 2i\epsilon^{\mu\nu\rho\sigma}\gamma_\sigma\gamma_5. \quad (27.2.5)$$

The term $-i\epsilon^{\mu\nu\rho\sigma}f_{\mu\nu}(\bar{\alpha}\gamma_\sigma\gamma_5\partial_\rho\lambda)$ does not contribute to $\int d^4x \delta\mathcal{L}$, because integrating by parts yields a contribution proportional to $\epsilon^{\mu\nu\rho\sigma}\partial_\rho f_{\mu\nu}$, which vanishes for an $f_{\mu\nu}$ of the form (27.2.1). This identity then allows us to rewrite the variation of the λ -term as

$$\delta(\bar{\lambda}\not{\partial}\lambda) = -f^{\mu\nu}(\bar{\alpha}(\gamma_\nu\partial_\mu - \gamma_\mu\partial_\nu)\lambda) + 2iD(\bar{\alpha}\gamma_5\not{\partial}\lambda).$$

The cancellation of terms proportional to $f^{\mu\nu}\lambda$ requires that $c_\lambda = 1/2$, while the cancellation of terms proportional to $D\lambda$ requires that $c_D = -c_\lambda$, so the supersymmetric Lagrangian density takes the form

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} - \frac{1}{2}(\bar{\lambda}\not{\partial}\lambda) + \frac{1}{2}D^2. \quad (27.2.6)$$

This shows that with V^μ canonically normalized, the field λ related to V^μ by the transformation rules (27.2.2) and (27.2.3) is also canonically normalized.

In addition, for Abelian gauge theories there is a superrenormalizable term, known as a *Fayet–Iliopoulos term* [2]:

$$\mathcal{L}_{\text{FI}} = \xi D, \quad (27.2.7)$$

with ξ an arbitrary constant. Its variation under a supersymmetry transformation is shown by Eq. (27.2.4) to be a derivative, so that it yields another supersymmetric term in the action. As we will see in Section 27.5, the presence of such a term can provide a mechanism for the spontaneous breakdown of supersymmetry.

Both for its own interest and as a tool in constructing supersymmetric interactions involving the fields $f_{\mu\nu}$, λ , and D , it is interesting to ask what sort of superfield has these as component fields. Somewhat surprisingly, it turns out to be a *spinor superfield* $W_\alpha(x)$, with component fields (in the notation of Eq. (26.2.10)) given by

$$\begin{aligned} C_{(\alpha)}(x) &= \lambda_\alpha(x), \\ \omega_{(\alpha)\beta}(x) &= \frac{i}{2}(\gamma^\mu\gamma^\nu\epsilon)_{\alpha\beta}f_{\mu\nu}(x) + (\gamma_5\epsilon)_{\alpha\beta}D(x), \\ V_{(\alpha)\mu}(x) &= -i\partial_\mu(\gamma_5\lambda(x))_\alpha, \\ M_{(\alpha)}(x) &= -i(\not{\partial}\gamma_5\lambda(x))_\alpha, \quad N_{(\alpha)}(x) = -(\not{\partial}\lambda(x))_\alpha, \\ \lambda_{(\alpha)\beta}(x) &= D_{(\alpha)}(x) = 0. \end{aligned} \quad (27.2.8)$$

(The subscript α on these component fields is put inside parentheses to emphasize that it labels the whole superfield.) It is straightforward to use Eqs. (27.2.2)–(27.2.4) to check directly that the superfield components given by Eq. (27.2.8) do transform as in Eqs. (26.2.11)–(26.2.17).

¹To derive this, use the fact that any 4×4 matrix may be expressed as a linear combination of the 16 independent covariant matrices described in Section 5.4, which in our case is limited by Lorentz invariance and space inversion invariance to the terms shown here. The coefficients of these terms can be calculated by giving $\mu\nu\rho$ the values 121 and 123.

Inserting the component fields (27.2.8) in Eq. (26.2.10), and using Eq. (26.A.5), we find that the superfield W_α takes the form

$$\begin{aligned} W_\alpha(x, \theta) = & \left[\lambda(x) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta f_{\mu\nu}(x) - i\gamma_5 \theta D(x) - \frac{1}{2} (\theta^T \epsilon \theta) \not{\partial} \gamma_5 \lambda(x) \right. \\ & + \frac{1}{2} (\theta^T \epsilon \gamma_5 \theta) \not{\partial} \lambda(x) + \frac{1}{2} (\theta^T \epsilon \gamma^\mu \theta) \gamma_5 \partial_\mu \lambda(x) \\ & - \frac{1}{4} (\theta^T \epsilon \theta) \gamma_5 \gamma^\mu \gamma^\nu \gamma^\sigma \theta \partial_\sigma f_{\mu\nu}(x) \\ & \left. + \frac{i}{2} (\theta^T \epsilon \theta) \gamma^\sigma \theta \partial_\sigma D(x) - \frac{1}{8} (\theta^T \epsilon \theta)^2 \square \lambda(x) \right]_\alpha. \end{aligned} \quad (27.2.9)$$

As we showed in Section 26.3, a superfield like this with zero λ - and D -components is *chiral*—that is, it is the sum of left-chiral and right-chiral superfields

$$W(x, \theta) = W_L(x, \theta) + W_R(x, \theta). \quad (27.2.10)$$

Here the left- and right-chiral superfields are simply the projections of W on the subspaces with $\gamma_5 = +1$ and $\gamma_5 = -1$, respectively:

$$\begin{aligned} W_L(x, \theta) &= \frac{1}{2} (1 + \gamma_5) W(x, \theta) \\ &= \lambda_L(x_+) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_L f_{\mu\nu}(x_+) + (\theta_L^T \epsilon \theta_L) \not{\partial} \lambda_R(x_+) - i\theta_L D(x_+). \end{aligned} \quad (27.2.11)$$

$$\begin{aligned} W_R(x, \theta) &= \frac{1}{2} (1 - \gamma_5) W(x, \theta) \\ &= \lambda_R(x_-) + \frac{1}{2} \gamma^\mu \gamma^\nu \theta_R f_{\mu\nu}(x_-) - (\theta_R^T \epsilon \theta_R) \not{\partial} \lambda_L(x_-) - i\theta_R D(x_-), \end{aligned} \quad (27.2.12)$$

with $x_\pm^\mu$ given by Eq. (26.3.23).

As we saw in Section 26.3, we can construct suitable Lagrangian densities from the $\mathcal{F}$ -term of any scalar function of a left-chiral superfield, plus its Hermitian adjoint. The simplest scalar function of the left-chiral superfield (27.2.11) is $\sum_{\alpha\beta} \epsilon_{\alpha\beta} W_{L\alpha} W_{L\beta}$. To calculate the $\mathcal{F}$ -term, we note that, when expressed as a function of θ_L and x_+ , the term in $\sum_{\alpha\beta} \epsilon_{\alpha\beta} W_{L\alpha} W_{L\beta}$ of second order in θ_L is

$$\begin{aligned} - \left[\sum_{\alpha\beta} \epsilon_{\alpha\beta} W_{L\alpha} W_{L\beta} \right]_{\theta_L^2} &= (\theta_L^T \epsilon \theta_L) \left[-2(\lambda_L^T(x) \epsilon \not{\partial} \lambda_R(x)) + D^2(x) \right] \\ &\quad + \frac{1}{16} (\overline{\theta_L} [\gamma^\mu, \gamma^\nu] [\gamma^\rho, \gamma^\sigma] \theta_L) f_{\mu\nu}(x) f_{\rho\sigma}(x). \end{aligned}$$

(The argument of the fields here can be taken as x^μ rather than x_+^μ because the difference would yield terms that contain at least three factors of θ_L , which therefore vanish.) Lorentz invariance together with the fact that $(\bar{s}[\gamma_\mu, \gamma_\nu]s)$ and $(\bar{s}[\gamma_\mu, \gamma_\nu]\gamma_5 s)$ vanish for any Majorana fermion s tell us that the bilinear $(\overline{\theta_L} [\gamma^\mu, \gamma^\nu] [\gamma^\rho, \gamma^\sigma] \theta_L)$ must be proportional to a linear combination of $(\overline{\theta_L} \theta_L)(\eta^{\mu\rho} \eta^{\nu\sigma} - \eta^{\mu\sigma} \eta^{\nu\rho})$ and $(\overline{\theta_L} \theta_L) \epsilon^{\mu\nu\rho\sigma}$. We can find the coefficients by giving $\mu\nu\rho\sigma$ the values 1212 or 1230, and in this way we find that

$$\begin{aligned} (\overline{\theta_L} [\gamma^\mu, \gamma^\nu] [\gamma^\rho, \gamma^\sigma] \theta_L) &= 4(\overline{\theta_L} \theta_L) [-\eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho} + i\epsilon^{\mu\nu\rho\sigma}]. \end{aligned}$$

The $\mathcal{F}$ -term is the coefficient of $(\bar{\theta}_L \theta_L)$, so

$$-\left[\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}\right]_{\mathcal{F}} = -2(\bar{\lambda}_R\not{\partial}\lambda_R) - \frac{1}{2}f_{\mu\nu}f^{\mu\nu} + \frac{i}{4}\epsilon^{\mu\nu\rho\sigma}f_{\mu\nu}f_{\rho\sigma} + D^2. \quad (27.2.13)$$

Eq. (26.A.21) shows that $(\bar{\lambda}\not{\partial}\lambda)$ is real, while $(\bar{\lambda}\not{\partial}\gamma_5\lambda)$ is imaginary, so the real part of Eq. (27.2.13) yields the Lagrangian (27.2.6) for the gauge and gaugino fields

$$-\frac{1}{2}\text{Re}\left[\sum_{\alpha\beta}\epsilon_{\alpha\beta}W_{L\alpha}W_{L\beta}\right]_{\mathcal{F}} = -\frac{1}{2}(\bar{\lambda}\not{\partial}\lambda) - \frac{1}{4}f_{\mu\nu}f^{\mu\nu} + \frac{1}{2}D^2. \quad (27.2.14)$$

The physical significance of the imaginary part will be discussed in a more general context in the next section.

There is another way to derive the form of the spinor superfield, which will turn out to provide a more convenient way of deriving the components of the gauge superfield in non-Abelian gauge theories. A tedious but straightforward calculation shows that the gauge-invariant superfield (27.2.9) may be expressed in terms of the gauge superfield (27.1.16) as

$$W_\alpha(x, \theta) = \frac{i}{4}(\mathcal{D}^T \epsilon \mathcal{D})\mathcal{D}_\alpha V(x, \theta), \quad (27.2.15)$$

where $\mathcal{D}_\alpha$ is the superderivative introduced in Eq. (26.2.26):

$$\mathcal{D}_\alpha \equiv \sum_{\beta}(\gamma_5 \epsilon)_{\alpha\beta} \frac{\partial}{\partial \theta_\beta} - (\gamma^\mu \theta)_\alpha \frac{\partial}{\partial x^\mu} = -\frac{\partial}{\partial \bar{\theta}_\alpha} - (\gamma^\mu \theta)_\alpha \frac{\partial}{\partial x^\mu}.$$

This result might have been obtained (aside from the normalization factor) by noting that the function (27.2.15) has the desired property of being a gauge-invariant chiral spinor superfield. First, note that Eq. (27.2.15) is a superfield, because it is formed by acting on the superfield V with superderivatives. Also, it follows from the anticommutation of the $\mathcal{D}$ s that the product of any three or more $\mathcal{D}_L$ s or three or more $\mathcal{D}_R$ s vanishes, so that

$$(\mathcal{D}^T \epsilon \mathcal{D})\mathcal{D} = (\mathcal{D}_L^T \epsilon \mathcal{D}_L)\mathcal{D}_R + (\mathcal{D}_R^T \epsilon \mathcal{D}_R)\mathcal{D}_L. \quad (27.2.16)$$

Because $\mathcal{D}_L(\mathcal{D}_L^T \epsilon \mathcal{D}_L) = \mathcal{D}_R(\mathcal{D}_R^T \epsilon \mathcal{D}_R) = 0$, the superfield (27.2.15) is chiral, with

$$\begin{aligned} W_{L\alpha}(x, \theta) &= \frac{i}{4}(\mathcal{D}_R^T \epsilon \mathcal{D}_R)\mathcal{D}_{L\alpha}V(x, \theta), \\ W_{R\alpha}(x, \theta) &= \frac{i}{4}(\mathcal{D}_L^T \epsilon \mathcal{D}_L)\mathcal{D}_{R\alpha}V(x, \theta). \end{aligned} \quad (27.2.17)$$

Finally, we can show that (27.2.15) is invariant under the generalized gauge transformation (27.1.13), which for a single Abelian gauge superfield is simply

$$V(x, \theta) \longrightarrow V(x, \theta) + \frac{i}{2}[\Omega(x, \theta) - \Omega^*(x, \theta)], \quad (27.2.18)$$

where $\Omega(x, \theta)$ is an arbitrary left-chiral superfield. Since $\mathcal{D}_L \Omega^* = 0$, the change in $W_{L\alpha}$ is proportional to $(\mathcal{D}_R^T \epsilon \mathcal{D}_R) \mathcal{D}_{L\alpha} \Omega$. But $\mathcal{D}_R \Omega = 0$ and

$$[(\mathcal{D}_R^T \epsilon \mathcal{D}_R), \mathcal{D}_{L\alpha}] = -2 [(1 + \gamma_5) \not{\partial} \mathcal{D}_R]_\alpha,$$

so the change in $W_{L\alpha}$ vanishes. A similar argument shows that $W_{R\alpha}$ is also gauge-invariant. (The work of checking Eq. (27.2.15) is greatly reduced by using this gauge-invariance property to put $V(x, \theta)$ in Wess–Zumino gauge.)

The chiral superfields (27.2.11) and (27.2.12) are evidently not of the most general form for left- and right-chiral superfields. To put the constraints satisfied by these superfields in a manifestly supersymmetric form, we note by using the anticommutation relation (26.2.30) that

$$\begin{aligned} \epsilon_{\alpha\beta} \mathcal{D}_{L\alpha} (\mathcal{D}_R^T \epsilon \mathcal{D}_R) \mathcal{D}_{L\beta} &= -2 \mathcal{D}_{R\alpha} \mathcal{D}_{L\beta} (\epsilon (1 + \gamma_5) \not{\partial})_{\beta\alpha} \\ &\quad + (\mathcal{D}_R^T \epsilon \mathcal{D}_R) (\mathcal{D}_L^T \epsilon \mathcal{D}_L) \\ &= \epsilon_{\alpha\beta} \mathcal{D}_{R\alpha} (\mathcal{D}_L^T \epsilon \mathcal{D}_L) \mathcal{D}_{R\beta}. \end{aligned} \tag{27.2.19}$$

From Eq. (27.2.17) it follows then that W_L and W_R are related by the constraint

$$\epsilon_{\alpha\beta} \mathcal{D}_{L\alpha} W_{L\beta} = \epsilon_{\alpha\beta} \mathcal{D}_{R\alpha} W_{R\beta}. \tag{27.2.20}$$

It is straightforward to show that the most general chiral spinor superfields satisfying Eq. (27.2.20) are of the form (27.2.11) and (27.2.12), with $f_{\mu\nu}$ constrained to satisfy the ‘Bianchi’ identities $\epsilon^{\mu\nu\rho\sigma} \partial_\rho f_{\mu\nu} = 0$.